

Q1.

Scientists sometimes replace one scientific model with a different model.

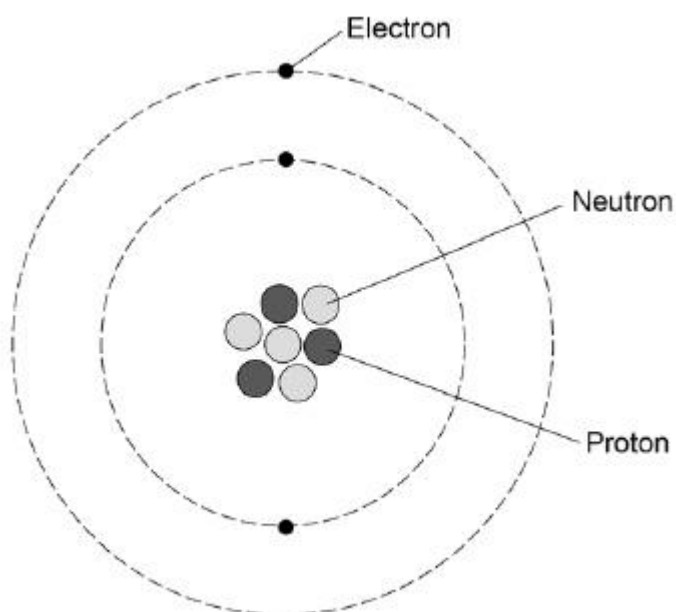
For example, in the early 20th Century the plum pudding model of the atom was replaced by the nuclear model of the atom.

Explain what led to the plum pudding model of the atom being replaced by the nuclear model of the atom.

(Total 6 marks)

Q2.

The diagram shows a lithium atom.



- (a) What is the mass number of this lithium atom?

Tick **one** box.

3 ☐

4 ☐

7 ☐

10 ☐

(1)

- (b) What is the atomic number of a lithium atom?

Tick **one** box.

3 ☐

4 ☐

7 ☐

10 ☐

Give a reason for your answer.

(2)

- (c) Complete the sentence.

Choose the answer from the box.

circles	levels	rings
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The electrons in an atom orbit in different energy _____.

(1)

- (d) Some atomic nuclei are unstable and decay by emitting an alpha particle or a beta particle.

Complete the symbols for an alpha particle and a beta particle.

Use answers from the box.

-1	0	1	2	4
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Alpha

☐	He
☐	

Beta

☐	⁰ e
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(3)

- (e) Doctors may use nuclear radiation to diagnose certain types of illness.

The table below gives data about three radiation sources used.

Each source emits beta radiation.

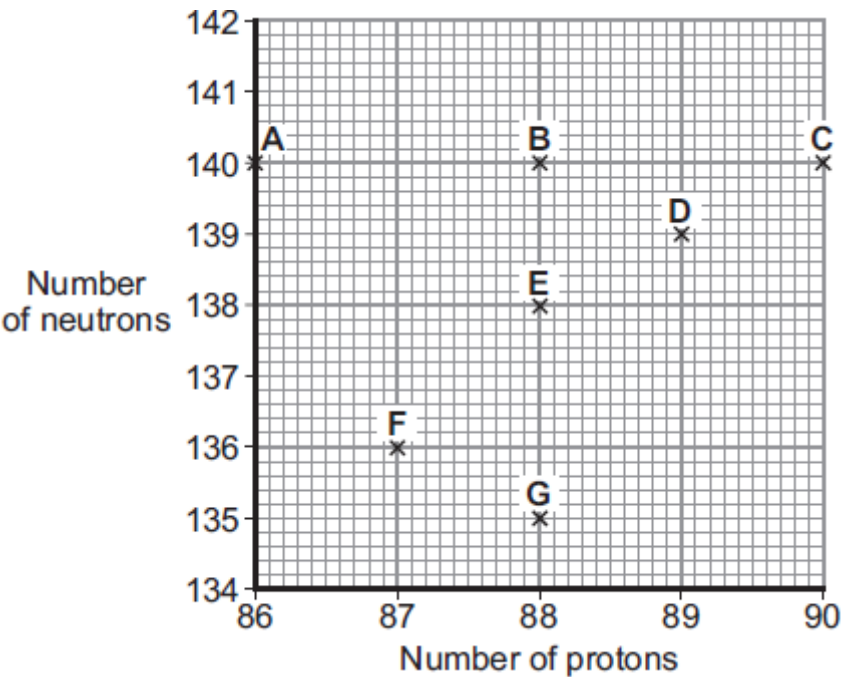
Radiation source	Half-life in minutes
Carbon-11	20
Nitrogen-13	10
Oxygen-15	2

Explain why oxygen-15 is likely to pose the least risk to a patient.

(2)
(Total 9 marks)

Q3.

- (a) The chart gives the number of protons and neutrons within the nuclei of 7 different atoms, **A – G**.



Which of these atoms are isotopes of the same element?

Give a reason for your answer.

(2)

- (b) Radium-226 is a radioactive isotope that decays into radon gas by emitting alpha particles.

The decay can be represented by the equation below.



- (i) Complete the equation by writing the correct number in each of the boxes.

(2)

- (ii) A sample of radium-226 has a count rate of 400 counts per second. The half-life of radium-226 is 1600 years.

How long will it be before the count rate has fallen to 50 counts per second?

Show clearly how you work out your answer.

Length of time = _____ years

(2)

- (c) In 1927, a group of women who had been employed to paint watch faces with a luminous paint sued their former employer over the illnesses caused by the paint. The women had been told that the paint, which contained radium, was harmless.

The company owners and the scientists working for the company knew that radium was harmful and took precautions to protect themselves from the radiation. The women were given no protection.

What important issue did the treatment of the women by the company owners and scientists raise?

Draw a ring around your answer.

economic environmental ethical social

Give a reason for your answer.

(2)

- (d) In the 1920s, many people, including doctors, thought that radium could be used as a treatment for a wide range of illnesses. Medical records that suggested radium

could be harmful were generally ignored. When some of the women who had used the luminous paint died, their deaths were not blamed on radium.

Suggest a reason why the evidence suggesting that radium was harmful was generally ignored.

(1)
(Total 9 marks)

Q4.

A teacher used a Geiger-Muller tube and counter to measure the number of counts in 60 seconds for a radioactive rock.

- (a) The counter recorded 819 counts in 60 seconds. The background radiation count rate was 0.30 counts per second.

Calculate the count rate for the rock.

Count rate = _____ per second

(3)

- (b) A householder is worried about the radiation emitted by the granite worktop in his kitchen.

1 kg of granite has an activity of 1250 Bq. The kitchen worktop has a mass of 180 kg.

Calculate the activity of the kitchen worktop in Bq.

Activity = _____ Bq

(2)

- (c) The average total radiation dose per year in the UK is 2.0 millisieverts.

The table below shows the effects of radiation dose on the human body.

Radiation dose in millisieverts	Effects
10 000	Immediate illness; death within a few weeks
1000	Radiation sickness; unlikely to cause death
100	Lowest dose with evidence of causing cancer

The average radiation dose from the granite worktop is 0.003 millisieverts per day.

Explain why the householder should **not** be concerned about his yearly radiation dose from the granite worktop.

One year is 365 days.

(2)

- (d) Bananas are a source of background radiation. Some people think that the unit of radiation dose should be changed from sieverts to Banana Equivalent Dose.

Suggest **one** reason why the Banana Equivalent Dose may help the public be more aware of radiation risks.

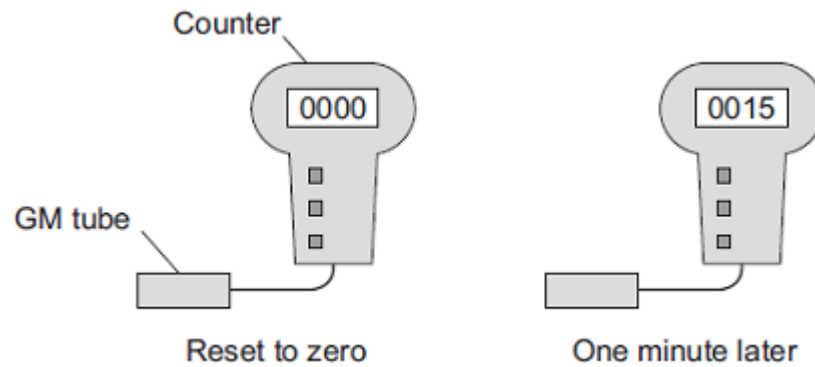
(1)

(Total 8 marks)

Q5.

- (a) A teacher used a Geiger-Müller (GM) tube and counter to measure the *background radiation* in her laboratory.

The teacher reset the counter to zero, waited one minute and then took the count reading. The teacher repeated the procedure two more times.



- (i) Background radiation can be either from natural sources or from man-made sources.

Name **one man-made** source of background radiation.

_____ (1)

- (ii) The three readings taken by the teacher are given in the table.

Count after one minute
15
24
18

The readings given in the table are correct.

Why are the readings different?

(1)

- (b) Some scientists say they have found evidence to show that people living in areas of high natural background radiation are less likely to develop cancer than people living in similar areas with lower background radiation.

The evidence these scientists found does not definitely mean that the level of background radiation determines whether a person will develop cancer.

Suggest a reason why.

(1)

- (c) An atom of the isotope radon-222 emits an alpha particle and decays into an atom of polonium.

An alpha particle is the same as a helium nucleus. The symbol below represents an alpha particle.



- (i) How many protons and how many neutrons are there in an alpha particle?

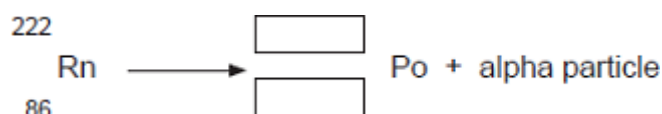
Number of protons = _____

Number of neutrons = _____

(2)

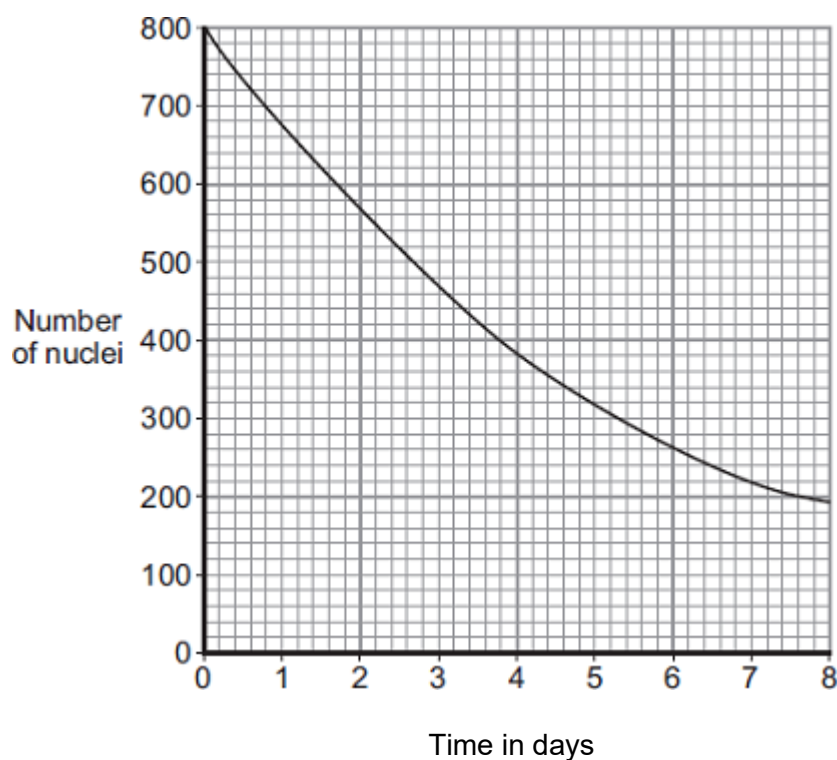
- (ii) The decay of radon-222 can be represented by the equation below.

Complete the equation by writing the correct number in each of the **two** boxes.



(2)

- (d) The graph shows how, in a sample of air, the number of radon-222 nuclei changes with time.



Use the graph to find the half-life of radon-222.

Show clearly on the graph how you obtain your answer.

Half-life = _____ days

(2)

(Total 9 marks)

Q6.

Radioactive waste from nuclear power stations is a man-made source of background radiation.

- (a) Which of the following is also a man-made source of background radiation?

Tick (✓) **one** box.

cosmic rays

☐

radiotherapy

☐

rocks

☐

stars

☐

(1)

- (b) Nuclear power stations use the process of nuclear fission.

Complete the sentences to describe the process of nuclear fission.

Choose answers from the box.

a neutron	a proton	an electron	
cosmic rays	energy	gamma rays	x-rays

An unstable nucleus absorbs _____ and splits into two parts.

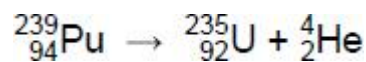
Two or three neutrons are released, as well as _____

and _____.

(3)

- (c) Plutonium-239 is one type of radioactive waste from nuclear power stations.

The following nuclear equation represents the decay of plutonium-239 (Pu-239).



How does the nuclear equation show that alpha radiation is emitted when plutonium-239 decays?

Tick (✓) **one** box.

An alpha particle contains 92 protons.

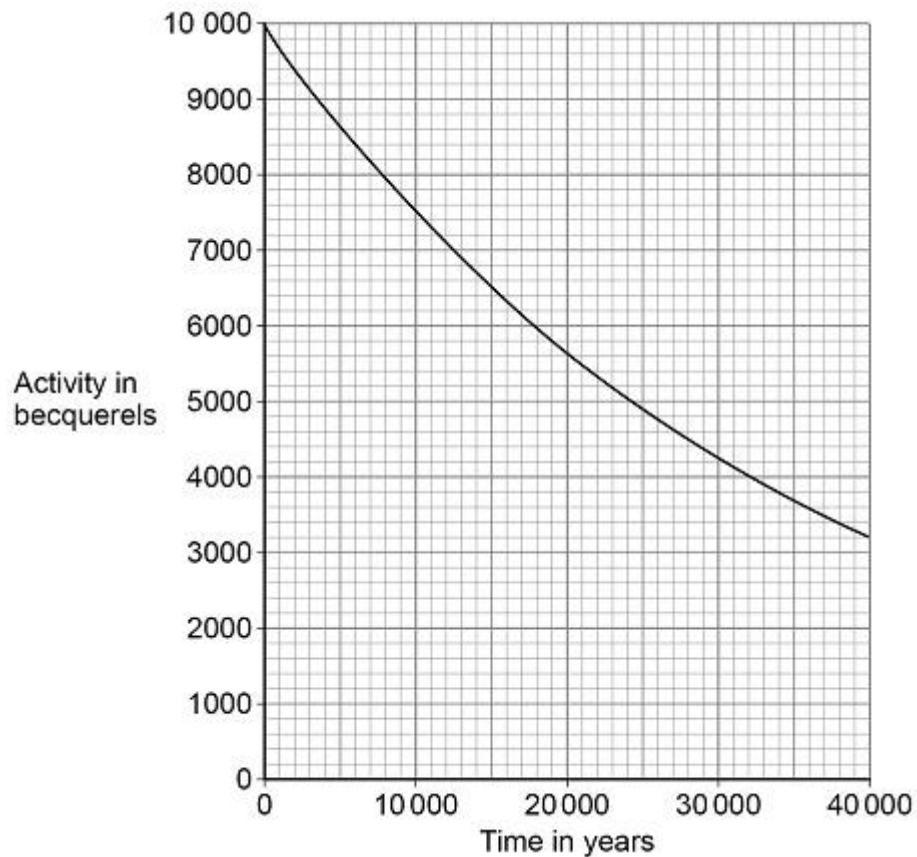
☐

An alpha particle has a mass number of 235.

An alpha particle is the same as a helium nucleus.

(1)

The graph below shows how the activity of a sample of plutonium-239 varies with time.



- (d) How much time will it take for the activity of the sample of plutonium-239 to fall to half of its initial activity?

Time = _____ years

(1)

- (e) What is the half-life of plutonium-239?

Half-life = _____ years

(1)

- (f) The radioactive waste from a nuclear power station is buried underground.

People are warned to stay away from places where radioactive waste is buried.

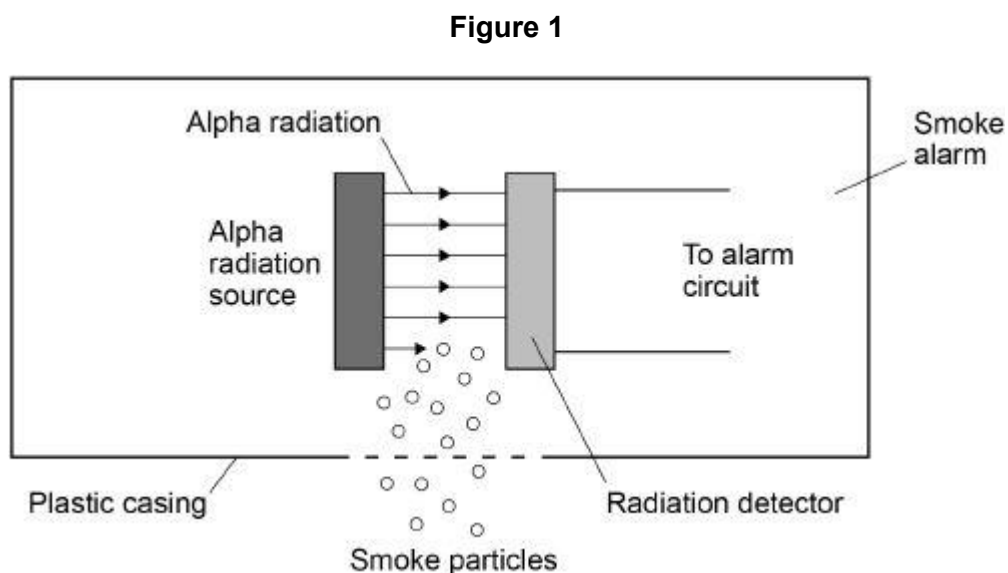
Suggest **one** risk of going near the place where radioactive waste is buried.

(1)
(Total 8 marks)

Q7.

Smoke alarms contain an alpha radiation source and a radiation detector.

Figure 1 shows part of the inside of a smoke alarm.



- (a) The smoke alarm stays off while alpha radiation reaches the detector.

Why does the alarm switch on when smoke particles enter the plastic casing?

(1)

- (b) Why is it safe to use a source of alpha radiation in a house?

(1)

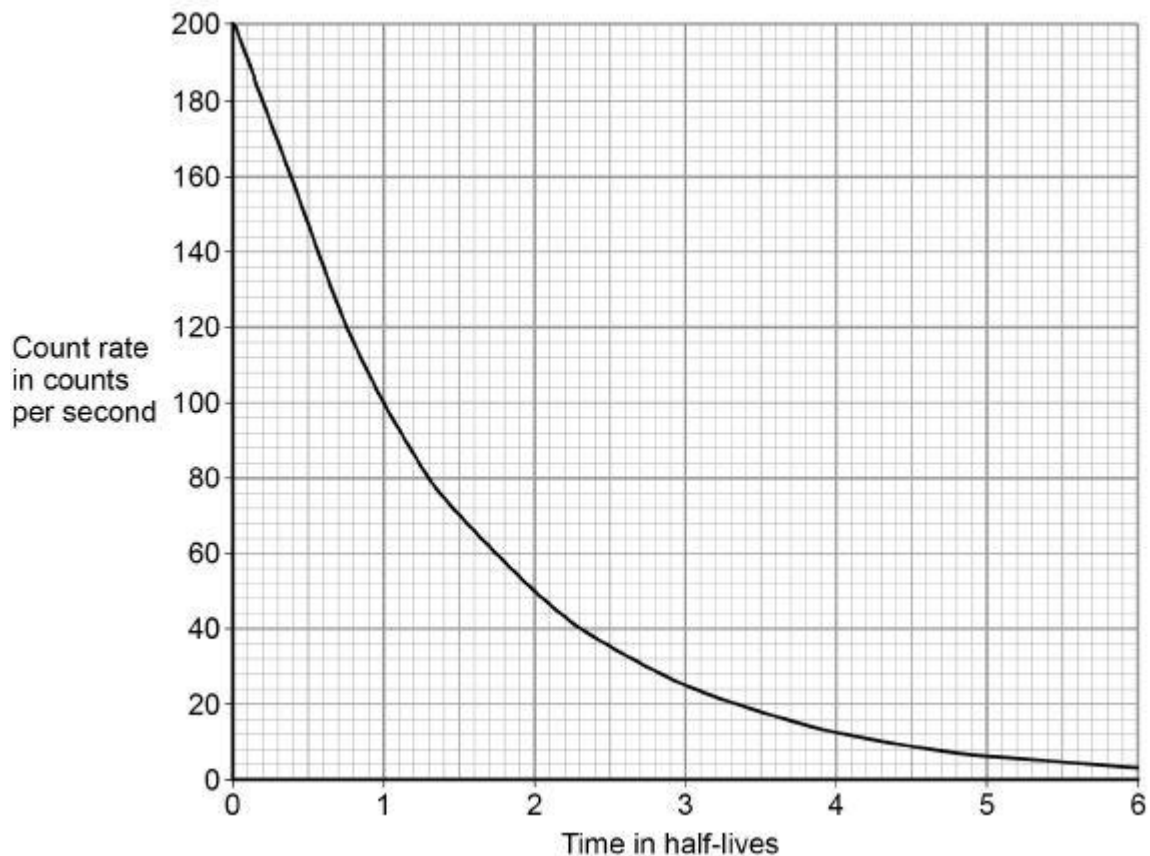
- (c) The smoke alarm would not work with a radiation source that emits beta or gamma radiation.

Explain why.

(2)

- (d) **Figure 2** shows how the count rate detected from the radiation source in the smoke alarm changes with time.

Figure 2



The smoke alarm switches on when the count rate falls to 80 counts per second.

Explain why the radiation source inside the smoke alarm should have a long half-life.

(2)

- (e) **Figure 3** shows a patient who has been injected with a radioactive source for medical diagnosis.

Figure 3

Radiation detector



Explain the ideal properties of a radioactive source for use in medical diagnosis.

(4)

(Total 10 marks)

Mark schemes

Q1.

Level 3 (5–6 marks):

A detailed and coherent explanation is provided. The student gives examples that argue a strong case and demonstrate deep knowledge. The student makes logical links between clearly identified, relevant points.

Level 2 (3–4 marks):

An attempt to link the description of the experiment and the results with differences between the two models. The student gives examples of where the plum pudding model does not explain observations. The logic used may not be clear.

Level 1 (1–2 marks):

Simple statements are made that the nuclear model is a better model. The response may fail to make logical links between the points raised.

0 marks:

No relevant content.

Indicative content

- alpha particle scattering experiment
- alpha particles directed at gold foil
- most alpha particles pass straight through
- (so) most of atom is empty space
- a few alpha particles deflected through large angles
- (so) mass is concentrated at centre of atom
- (and) nucleus is (positively) charged
- plum pudding model has mass spread throughout atom
- plum pudding model has charge spread throughout atom

[6]

Q2.

- | | | |
|-----|---------------------------------------|---|
| (a) | 7 | 1 |
| (b) | 3 | 1 |
| | number of <u>protons</u> | |
| | <i>reason only scores if 3 chosen</i> | 1 |
| (c) | levels | 1 |
| (d) | ${}^4_2\text{He}$ | 1 |
| | <i>correct order only</i> | 1 |
| | ${}^0_{-1}\text{e}$ | 1 |

(e) shorter half-life (than the other sources)

1

exposure time to radiation is shorter

1

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Q3.

(a) **B E G**

*all 3 required and no other
any order*

1

same number of / 88 protons (and different numbers of neutrons)
same number of electrons is insufficient

1

(b) (i) 222

1

86

1

(ii) 4800

allow 1 mark for obtaining 3 half-lives

2

(c) ethical

1

deceived / lied to (about safety of working conditions)
*accept (women) not warned of the dangers
given no protection is insufficient*

or

value own / scientists' lives more than women

or

did not treat women humanely

1

(d) accept any sensible suggestion

eg

too many interests in continued use of radium

evidence may cause public unrest

*do **not** accept not enough evidence*

doctors not want to be blamed for illnesses (caused by radium)

*accept doctors not wanting to be sued (for harm caused by
using radium)*

doctors thought (possible) benefits outweighed (possible) risks

*do **not** accept did not know radium could be harmful
believe radium could treat illnesses is insufficient*

1

[9]

Q4.

(a) $\text{count rate} = \frac{819}{60}$ 1

count rate = 13.65 1

corrected count rate = 13.35 (per second)

allow an answer of

*background = 0.30×60
= 18 (per minute)*

*corrected count rate
= $819 - 18$*

*corrected count rate
= 801 per minute*

1

an answer of 13.35 (per second) scores 3 marks

an answer of 13.95 (per second) scores 2 marks

an answer of 801 (per second) scores 2 marks

(b) activity = 1250×180 1

activity = 225 000 (Bq) 1

an answer of 225 000 (Bq) scores 2 marks

(c) yearly dose = 0.003×365
allow yearly dose = 1.095 (mSv) 1

which is $\ll 100$ (mSv)

or

(well) below the lowest dose with evidence of causing cancer / harm 1

(d) people are able to compare a radiation risk / dose / hazard to the radiation dose from (eating) bananas 1

[8]

Q5.

(a) (i) any **one** from:

- nuclear power (stations)
accept nuclear waste
accept coal power stations
- nuclear weapons (testing)
accept nuclear bombs / fallout
- nuclear accidents
accept named accident, eg Chernobyl or Fukushima

	accept named medical procedure which involves a radioactive source		
	accept radiotherapy		
	accept X-rays		
	accept specific industrial examples that involve a radioactive source		
	nuclear activity / radiation is insufficient		
	smoke detectors is insufficient		1
(ii)	(radioactive decay) is a random process		
	accept an answer in terms of background / radiation varies (from one point in time to another)		1
(b)	any one from:		
	• (maybe) other factors involved		
	accept a named 'sensible' factor, eg smoking		
	• evidence may not be valid		
	accept not enough data		
	• may not have (a complete) understanding of the process (involved)		1
(c)	(i)	2	1
		2	1
	(ii)	218	
		correct order only	1
		84	1
(d)	3.8 (days)		
	allow 1 mark for showing correct method using the graph provided no subsequent steps		
	correct answers obtained using numbers other than 800 and 400 gain 2 marks provided the method is shown		2
			[9]
Q6.			
(a)	radiotherapy		1
(b)	a neutron		1
	energy		
	energy and gamma rays can score in reverse order		

- gamma rays 1
- (c) An alpha particle is the same as a helium nucleus. 1
- (d) 24 000 (years) 1
allow an answer between 24 000 and 24 500 (years) inclusive
- (e) 24 000 (years) 1
or
 their (d)
- (f) Any **one** from: 1
 • irradiation
 • cancer
 • genetic damage
 • mutations to DNA / genes
 • radiation sickness / poisoning

[8]

Q7.

- (a) smoke absorbs / stops alpha radiation 1
allow alpha particles for alpha radiation
alpha radiation does not reach the detector is insufficient
- (b) alpha radiation is not very penetrating 1
allow alpha particles for alpha radiation
or
 alpha radiation does not penetrate skin
allow alpha radiation does not travel very far (in air)
- (c) beta and gamma radiation will penetrate smoke 1
allow beta and gamma radiation will not be stopped by smoke
 no change (in the count rate) would be detected
allow the change detected (in the count rate) would be too small
- (d) (a long half-life means) the count rate is (approximately) constant 1
allow activity of source is (approximately) constant
or
 a short half-life means the count rate decreases quickly
 until 1.3 half-lives the count rate is above 80 per second

allow after 1.3 half-lives the count rate is below 80 per second

or

until 1.3 half-lives the count rate is above the threshold for the smoke alarm to be activated

or

after 1.3 half-lives the smoke alarm will be activated all the time

so don't have to replace source or smoke detector is insufficient

1

- (e) **Level 2:** Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.

3–4

Level 1: Relevant points (reasons / causes) are identified, and there are attempts at logically linking. The resulting account is not fully clear.

1–2

No relevant content

0

Indicative content

- short half-life or half-life of a few hours
- (short half-life means) less damage to cells / tissues / organs / body
- low ionising power
- (low ionising power means) less damage to cells / tissues / organs / body
- highly penetrating
- (highly penetrating means) it can be detected outside the body
- emits gamma radiation

[10]

Examiner reports

Q3.

- (a) This question was quite well answered.
- (b)
 - (i) Just over 33% of students gained at least one mark.
 - (ii) Whilst many students had some idea that they were supposed to halve a number, a large number of them halved either the half-life or the nucleon number of 226. Another common mistake was to double the time as the count rate halved. Just over 55% of students scored zero.
- (c) Whilst about half of the students correctly chose 'ethical', few were able to give a reason, often just quoting the information given in the question that the women were given no protection.
- (d) Despite the question referring to 'the evidence suggesting that radium was harmful', a large number of answers stated that there was 'no evidence'.

Q4.

Foundation

- (a) Around 9% of students did not attempt this question. Others randomly multiplied or divided the figures given, with no logic to their method.
Many students did not know how to deal with the background count rate; some ignored it altogether, others added it to the measured count rate, some divided the measure count rate by the background count rate. However, around 13% were able to subtract it to score all three marks.
Some students did the calculation as a rate of per minute, scoring two marks for an answer of 801, but failing to change the answer line unit to 'per minute'.
- (b) Nearly three quarters of students answered correctly, scoring two marks. Some scored one mark for showing that they were multiplying the two numbers together, but then making a mistake in the calculation. Of the few who did not gain any credit, the common errors were dividing the two given numbers or converting the mass into grams. Around 8% of students did not attempt this question. Students should be reassured that marks are not deducted for incorrect answers, so it is worth them attempting a simple calculation such as multiplying the two values given.
- (c) Around three quarters of students calculated the yearly dose from the granite worktop correctly, thus scoring one mark. Some ignored the values of dose per day and the number of days in a year which were given.
Around one quarter of students went on to compare their value to the lowest dose with evidence of causing cancer to explain why the householder need not be concerned, scoring the second mark. Some students failed to score this mark because they did not use information from the table.
Some related their calculated value to the average yearly radiation dose, which was insufficient, or calculated how many years they would need to be exposed to the worktop to suffer harm, which was also insufficient.
A few answers were seen where the yearly dose was calculated as 1 095 mSv (instead of 1.095 mSv). Students then went on to say that this would only cause radiation sickness, so the householder did not need to be concerned as it was unlikely to cause death.

- (d) Around 8% of students scored the mark for this question. Answers generally fell into one of several categories. One category involved an answer along the lines of the public needing to limit their intake of bananas because of the radiation which bananas emit. Another category gave the idea that the public would become aware that radiation was given off by everyday objects such as bananas. Another category included answers which stated that everyone knows what a banana is. The mark was awarded for answers indicating that it would be more meaningful to compare a certain dose of radiation with the number of bananas which would give the same dose. Unfortunately, some students who had the correct idea were often unable to express this idea clearly enough to gain credit.

Higher

- (a) 54% of students scored all 3 marks. Common mistakes were to add the value of 0.30 or multiply or divide by 0.30. These students scored 2 marks. Some students calculated 801 (counts per minute) but without changing the unit on the answer line, so only scored 2 marks.
- (b) 92% of students scored 2 marks for this question. Incorrect answers usually resulted from dividing the two figures given in the question.
- (c) 58% of students scored 2 marks. Students failing to score both marks usually did the calculation, but related it to the yearly dose rather than the dose linked with causing cancer.
- (d) 20% of students answered this question correctly. 'Everyone knows what a banana is', was a common answer, which was insufficient. The idea that a radiation dose could be compared with a number of bananas was enough to be creditworthy, although this could also be related to a risk or hazard.

Q5.

- (a) (i) Surprisingly only just over half of students gained this mark, with the popular answers being 'nuclear power stations' or 'nuclear weapons testing'. 'X-Rays' was also a common response. The common incorrect responses were smoke detectors and vague answers referring to medical uses without being specific. A significant number of students failed to see that 'man-made sources' was in bold and responded with 'cosmic rays' and 'rocks'.
- (ii) Several students gave the correct response in terms of background radiation varying but relatively few gave the answer in terms of decay being a random process. Common incorrect answers referred to the teacher moving the experiment around the lab to cause the differences or to other radioactive sources being present at the times of the three readings, including the introduction of radioactive materials into the room and the presence of different food being present when the readings were taken. Many did use 'vary' or 'changes' but too many gave responses showing a complete lack of understanding of this process.
- (b) The most common correct response was 'other factors'. When other factors were mentioned, it was usually smoking. Several students gave answers referring to the hereditary aspect of cancer. Very few answers talked about the validity of the evidence, or the sample size.
- (c) The majority of the students gained both marks.

- (d) Almost two thirds of the students gave the correct response of 3.8 days, with a few giving the answer as a fraction and one or two converting 0.8 days into hours and minutes. Nearly all correct answers used the counts of 800 and 400 to determine the half-life. A few misread the x axis scale and then answered 3.4 days and a few, after drawing correct lines then put down an incorrect answer. Common errors included believing the “222” of radon was important and so a line was drawn across from a count of 222 to give a half-life near 7 days.

Q7.

- (a) 57% of students scored 2 marks for this question, with 32% scoring 1 mark. The most common mistake was not stating which quantity should remain constant, ‘charge’ being a commonly seen incorrect answer.
- (b) 74% of students scored a mark for this question, usually for the idea of low penetrating ability. ‘Doesn’t penetrate skin’ was creditworthy, ‘doesn’t penetrate paper’ was insufficient. ‘Doesn’t penetrate the case’ was allowed.
- (c) 53% of students scored 1 mark for this question but very few scored 2 marks. Most students answered in terms of beta and gamma radiation being able to penetrate the smoke, but didn’t link this idea to the need for a change in the radiation reaching the detector for the alarm to sound.
- (d) Very few 2 mark answers were seen. Most students scored zero, with insufficient responses having to do with saving money, or not having to replace the source very often. Very few linked the graph to the question, and realised that 1.3 half-lives could be any amount of time depending on the radioisotope used.
- (e) Well answered by many students, 36% scoring 3 or 4 marks. Those who didn’t usually gave opposite statements to the ideal properties, such as low penetrating ability but highly ionising, which will have limited their overall score. The minimum required for a 4 mark answer would have been two properties with explanations, or four simple statements. Many students gave much fuller answers than this, however.